

Economia Para Engenharia $\Rightarrow$ Aula 06, Exercícios $\Rightarrow$ Victor C. de. Gouveia

$$1) i_m = \frac{0,06}{12} = 0,005 = 0,5\% \text{ a.m.}$$

$$M = P(1+i_m)^n \Rightarrow M = 1000 \cdot (1+0,005)^{48} \approx 12.705,,$$

$$2) V_f = V_p \cdot (1+i)^n \Rightarrow V_f = 5000 \cdot (1+0,12)^6 \approx 9456,60$$

$$3) V_f = V_p \cdot (1+i)^n \Rightarrow \frac{V_f}{V_p} = (1+i)^n \Rightarrow \sqrt[n]{\frac{V_f}{V_p}} = 1+i \Rightarrow i = \sqrt[n]{\frac{V_f}{V_p}} - 1$$

$$\Rightarrow i = \sqrt[4]{\left(\frac{7500}{500}\right)} - 1 = \sqrt[4]{1,5} - 1 \approx 1,1067 - 1 = 0,1067 = 10,67\% \text{ a.m.}$$

$$4) V_p = \frac{V_f}{(1+i)^n} \Rightarrow V_p = \frac{30.000}{(1+0,015)^{24}} \approx 20.335,20 \text{ reais,,}$$

$$5) V_{f_{\text{real}}} = 30000(1+0,06)^2 \approx 33708, V_p = \frac{33.708}{(1+0,015)^{14}} \approx 23593,$$

$$6) V_f = V_p(1+i)^n \Rightarrow m = \frac{75}{30} = 2,5 \text{ meses} \Rightarrow i = \sqrt[n]{\frac{V_f}{V_p}} - 1 = \sqrt[2,5]{\frac{12433}{3564}} - 1 \approx 1,0742 - 1 = 0,0742$$

7,42% a.m.

$$7) 1000 \cdot 9,34 = 9340; T.C. = 540,0002 \cdot 9340 = 541,334 = 6,934 = 6,33$$

$$C.L. = 3370 + 6,33 = 3376,33;$$

$$1000 \cdot 9,68 = 9680 \Rightarrow T.C. = 541,336 = 6,34 \Rightarrow V.L. = 3680 - 6,34 = 3673,66$$

$$\text{Resíduo} = 3673,66 - 3376,33 = 297,33; \text{ Taxa de retorno: } r = \frac{\text{resultado}}{\text{Compra líquida (C.L.)}}$$

$$r = \frac{297,33}{3376,33} \approx 0,08806 \approx 8,81\%.$$

$$8) a) i = \frac{0,40}{12} = 0,0333 \approx 3,33\% \text{ a.m.} \Rightarrow (1+0,0333)^{12} - 1 = \text{Taxa Anual} = 42,1\%$$

$$b) (1,02)^{12} - 1 \approx 12,68\% \text{ a.a.}; (1+0,3)^4 - 1 \approx 12,55\% \text{ a.a.}$$

$$T_{\text{total Anual}} = (1,1268) \cdot (1,1255) \cdot 1 \approx 26,18\%$$

$$g) V_f = \frac{V_f}{(1+i)^n} \Rightarrow V_p = \frac{2000}{(1+0,1)^3} = \frac{2000}{1,331} \approx 1471,68$$

$$D = V_f - V_p = 2000 - 1471,68 = 272,32\%$$

$$10) V_f = 5000 \cdot (1+0,05)^{7/11} = 5000 \cdot 1,03658333 \approx 5182,90\%$$

$$D = V_f - C = 5183 - 5000 = 25\%$$